

Lynbrook Math Summer Camp 2026

C2 Counting Techniques Part 1

Lynbrook Math Club

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C2.1 Lecture

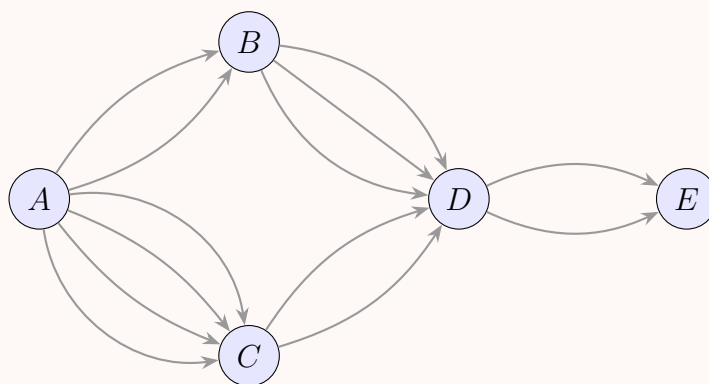
What happens when problems become too complex for directly applying permutations and combinations? Over the next few lessons, we will learn different creative techniques to solve more complicated problems.

Casework

When a problem feels too large or complicated to solve directly, it helps to break it into smaller, more manageable **cases**. We can solve each of these cases, and put them together at the end.

Example 1

Assuming you can only move in the direction of the arrows, how many ways can you get from Point A to point E ?



We see that the multiplication principle would be a good way to solve this problem, however, there's no way to directly apply it in one step. Instead, we have to break the problem into cases.

Let's start from A . From A , there are two ways to progress, going to B or going to C .

Case 1: Traveling via Point B ($A \rightarrow B \rightarrow D \rightarrow E$) By the multiplication principle, the number of ways to complete this route is the product of the number of paths between each consecutive stage:

- Number of paths from A to B : 2
- Number of paths from B to D : 3

- Number of paths from D to E : 2

Thus, the number of paths via B is:

$$2 \times 3 \times 2 = 12 \text{ ways}$$

Case 2: Traveling via Point C ($A \rightarrow C \rightarrow D \rightarrow E$) Similarly, we multiply the number of paths for each stage of this route:

- Number of paths from A to C : 4
- Number of paths from C to D : 2
- Number of paths from D to E : 2

Thus, the number of paths via C is:

$$4 \times 2 \times 2 = 16 \text{ ways}$$

Total Paths:

Since these two cases are mutually exclusive (a path cannot go through both B and C), we use the addition principle to find the total number of ways to travel from A to E :

$$\text{Total Ways} = 12 + 16 = \mathbf{28}$$

Notice that when we have a series of independent choices, we multiply the number of options in each step. However, when we have exclusive options (we can't choose more than one), we add the number of options.

Example 2

At an ice cream shop, there is a selection of five toppings. If a customer can select any number of toppings, from no toppings to all five toppings, how many combinations of toppings can the customer order?

We can do casework based on the number of toppings chosen.

Case 1: No toppings chosen

In this case, there is $\binom{5}{0} = 1$ way to choose the topping.

Case 2: 1 topping chosen

There are $\binom{5}{1} = 5$ ways to choose the topping.

Case 3: 2 toppings chosen

There are $\binom{5}{2} = 10$ ways to choose the toppings.

Case 4: 3 toppings chosen

There are $\binom{5}{3} = 10$ ways to choose the toppings.

Case 5: 4 toppings chosen

There are $\binom{5}{4} = 5$ ways to choose the toppings.

Case 6: 5 toppings chosen

There is $\binom{5}{5} = 1$ way to choose all five toppings,

Total combinations:

The total number of ways to choose the toppings is

$$1 + 5 + 10 + 10 + 5 + 1 = \mathbf{32}.$$

Do you notice anything interesting about the answer? There are 5 possible toppings to choose and the answer is 2^5 .

Claim C2.1.3 (A Binomial Identity) — For $n \geq 0$,

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n-1} + \binom{n}{n} = 2^n.$$

We can prove this combinatorially by equating the left-hand side with the total number of ways to form a committee that can include any number of people from 0 to n . Another way to select a committee is to decide if a person is or is not on the committee for each of the n people. Because there are n people and they can each be or not be on the committee, there are 2^n ways to select the committee. Since both ways to select the committee should yield the same number of total committees, $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n-1} + \binom{n}{n} = 2^n$.

Applying this to our ice cream problem, we can solve the problem differently from our original casework method by deciding whether or not to include each of the five toppings on our ice cream.

Complementary Counting

Sometimes, counting things we want can be challenging, while counting things we don't want is easier. In these scenarios, we can count the total number of things we don't want and subtract it from the total number of things to find the number of things we want. This technique is called complementary counting.

Example 4

I flip 5 fair coins. How many ways are there for me to flip the coins if I want at least one head?

Doing casework for this problem would be pretty tedious. We would need to have a case for rolling one head, rolling two heads, etc. However, finding the number of ways to roll less than one head is relatively easy. Therefore, we can use complementary counting.

The total number of ways to roll a die is $2^5 = 32$, and the total number of ways to roll zero heads, or all tails, is 1. Therefore, the number of ways to roll greater than zero, or at least one head is $32 - 1 = \mathbf{31}$.

Example 5

Sohum has 3 shirts, 3 pairs of pants, and 3 pairs of shoes, and each item of clothing is one of three colors, where he has no two of the same color of any item. How many ways can Sohumi choose an outfit if he doesn't want to wear the same color shirt, pants, and shoes?

We don't want to have to do casework for each possible combination, but we notice that the number of ways to choose an outfit that Sohumi wouldn't want is easier to find than the number of ways to choose an outfit Sohumi wants.

The number of total outfits we can put together is $3 \cdot 3 \cdot 3 = 27$. The number of outfits that are all one color is 3, because there are three colors to make all the items. Therefore, the number of outfits Sohumi would be satisfied with is $27 - 3 = \mathbf{24}$.

Something useful to keep in mind is to look out for questions that include *at least* or *at most*. These questions are often candidates to use complementary counting on.

Overcounting

Another way to make your life easier by not counting what you want directly is overcounting. If counting more than what we want is easier, we can deliberately count more, and then apply different operations to get what we want.

Example 6 (Repeated Items)

How many ways are there to arrange the letters in LYNBROOK?

We already know how to do this problem if we consider the two O's distinct (for example O_1 and O_2). The number of ways to arrange the 8 letters is $8! = 40320$. When we consider the two O's distinct, every valid word is counted twice (for instance, $LYNBRO_1O_2K$ and $LYNBRO_2O_1K$ are treated as separate arrangements). To find the number of arrangements when the O's are identical, we must divide the total $8!$ by 2 to get 20160.

In general, when arranging n items where some items are repeated, we handle overcounting by dividing by the factorial of each repeated group's count.

Claim C2.1.7 — If a set of n total objects contains k_1 identical objects of type 1, k_2 identical objects of type 2, and so on, the number of distinct permutations is:

$$\frac{n!}{k_1! \cdot k_2! \cdots k_m!}.$$

First, we can temporarily assume that all n objects are distinguishable (e.g. by labeling the k_1 objects of type 1 as $1_1, 1_2, \dots, 1_{k_1}$, and so on). There are $n!$ ways to arrange these n distinct objects in a row.

In reality, the k_1 objects of type 1 are identical to each other. For any specific layout of the n positions:

- The k_1 identical objects can be permuted among their k_1 positions in $k_1!$ ways without changing the visual arrangement.
- Similarly, the k_2 identical objects can be permuted among themselves in $k_2!$ ways.
- In general, for each type i , the k_i identical objects can be permuted among themselves in $k_i!$ ways.

Since these permutations within each group of identical objects are independent, every distinct visual arrangement of all n objects is counted exactly

$$k_1! \cdot k_2! \cdots k_m!$$

times in the initial $n!$ total permutations.

To obtain the number of unique arrangements, we divide the total permutations by the overcounting factor:

$$\text{Distinct Permutations} = \frac{n!}{k_1! \cdot k_2! \cdots k_m!}.$$

Example 8 (Repeated Items)

I want to arrange my 6 marbles in a line on my windowsill. I have 3 blue marbles, 2 green marbles, and 1 white marble. In how many ways can I arrange the marbles if colors of the same color are indistinguishable?

One way we can solve this problem is using combinations. However, we can also solve this using the technique from the last problem.

There are $6!$ ways to arrange all the marbles if we do not account for the repetition of the same color. To compensate for the overcounting, we must divide by the factorials of the repetitions for each color: $3!$ for the blue marbles, $2!$ for the green marbles, and $1!$ for the white marble. Thus, the total number of arrangements using this method is:

$$\frac{6!}{3! \cdot 2! \cdot 1!} = \frac{720}{6 \cdot 2 \cdot 1} = 60.$$

So far, we've mainly been working with arranging items in a line. What if we need to arrange things in a circle? This introduces rotational symmetry.

Example 9 (Rotational Symmetry)

There are five people who want to eat together at a round lunch table. How many ways are there to sit if two arrangements are considered the same if one can be achieved by rotating another?

If the people were sitting in a row instead of in a circle, there would be $5!$ arrangements. However, different row arrangements can correspond to the same circular arrangement. For example, the row arrangements

- A, B, C, D, E
- B, C, D, E, A
- C, D, E, A, B
- D, E, A, B, C
- E, A, B, C, D

give the same circular arrangement. We can see that five row arrangements will correspond to a single circular arrangement, so the total number of circular arrangements is $\frac{5!}{5} = 24$.

Another way to think about this problem is to fix one person at the "head" of the table and count the number of ways the other people can be arranged after fixing the head. Fixing one person breaks the rotational symmetry. Since the table is circular, whichever seat the first person picks acts as the anchor. It doesn't matter which chair they sit in because all seats are relative to one another before anyone sits down. However, once the first person is fixed, the remaining four positions relative to them are distinct. The remaining 4 people can be arranged in these distinct positions in $4! = 24$ ways.

Example 10 (Rotational Symmetry)

A group of six friends including Catherine and Jessica are having a dinner together. How many ways are there for the friends to sit at a round table if

- (a) there are no constraints on seating
- (b) Catherine and Jessica want to sit together
- (c) Catherine and Jessica want to sit across from each other
- (d) Catherine and Jessica don't want to sit together

- (a) We can fix one person's seat, and the number of ways to arrange the other people is $5! = 120$.
- (b) We can treat Catherine and Jessica as a single block, and proceed as usual. There are now five people (because Catherine and Jessica count as one person). We can fix one person's seat, and the number of ways to arrange the other people is $4! = 24$. However, within the Catherine and Jessica block, there are two ways to arrange them, so we need to multiply the number of arrangements by two to get 48 seatings.
- (c) If we fix Catherine's spot, Jessica's spot will also be fixed. Therefore, we can just fix their spots and find the number of ways to arrange the other four people, which is $4! = 24$.
- (d) We notice that we already know the number of seatings where Catherine and Jessica are together and the total number of seatings. To find the number of seatings where Catherine and Jessica are separated, we can use complementary counting and subtract the number of seatings where they are together from the total number of seatings to get $120 - 48 = 72$ seatings.

What if we add reflections?

Example 11 (Reflectional Symmetry)

Ellie is making a bracelet with 4 beads of different colors. If two possible bracelets are the same if they are equivalent through rotation, or reflection, how many different bracelets can she make?

We now know how to solve the problem if reflections isn't an issue. We just fix one of the beads and find the number of ways to arrange the other ones to get $3!$.

However, a bracelet is a 3-D object, every arrangement is equivalent to its mirror image. Since the beads are distinct, no arrangement is symmetric under reflection, so flipping always pairs up arrangements into identical pairs. Therefore, we need to divide by 2 to account for the reflectional symmetry to get $\frac{3!}{2} = 3$.

So far, we've dealt with overcounting by dividing. When every unique setup gets counted k times, division cleans it up nicely. But what happens when sets overlap in more complex ways? In the next lesson, we'll look at another technique, the Principle of Inclusion-Exclusion (PIE), where we fix overcounting by addition and subtraction.